The Rise and Fall of Ja Morant

A Statistical Account, 2017–2026

GRAVITY Research Note No. 1

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Abstract

We measure every season of Ja Morant's career — college, rise, peak, and collapse — with one rating system (GRAVITY) applied the same way to every player in every season, so his trajectory can be read in a single currency. Three findings organize the paper. **(1) The rise was real:** from an average-ish rookie to the 97th percentile of the league in 2021–22. **(2) The collapse was about missing, not choosing:** measured across shot zones, 88% of his 2025–26 efficiency crash came from worse conversion and only 12% from where he shot — while his playmaking hit a career high. **(3) Much of the "decline" is too small a sample to judge:** the three-point collapse everyone saw covers just 85 attempts — too few to distinguish genuine decline from ordinary shooting variance — and his free-throw shooting, the purest touch test, actually *improved*, significantly. The fall of Ja Morant is, by the numbers, mostly an *availability* story: 79 games played in three seasons. His rank of #140 of 644 prices that risk, and the number is robust to the model's own judgment calls: every variation we test in Appendix B lands him between #112 and #147. It probably underprices the talent, and we show exactly where the difference lives.

1 Introduction

Between March 2019 and March 2022, Ja Morant went from unranked recruit to the second pick of the draft, Rookie of the Year, Most Improved Player, All-NBA, and the face of a 56-win team. Between March 2022 and June 2026 he collected two firearm suspensions (33 games), a season-ending shoulder surgery, a season-ending elbow injury, a one-game team suspension, and finally a trade to Portland for two role players.

Everyone has a story about this arc. This paper has measurements. We run every Morant season through GRAVITY, the player-rating system behind our July 2026 ranking of all 644 NBA players, and we show the arithmetic at every step. Where the numbers can't support a firm conclusion — and in a 569-minute season they often can't — we say so with confidence intervals instead of adjectives.

2 How the rating works (the short version)

GRAVITY asks two questions about a player. *How good is he per minute on the floor?* — measured by blending box-score impact, scoring efficiency relative to his offensive burden, playmaking, defense, and team on/off data (garbage-time filtered, from Cleaning the Glass). *And how much can you actually count on him?* — measured by games played, role size, and documented injuries. The two multiply. A per-minute star who misses most of the season lands well below his talent level, on purpose.

Everything is expressed in **z-scores**: how many standard deviations a player sits above or below the league-average minute. As a reading guide: 0 is league average, +1 is clearly good, +2 is elite, +3 is historic; negative is below average. The final score is put on a friendly scale,

$$G \;=\; 50 + 15 \times \big(\text{per-minute impact, shrunk for small samples}\big) \times \big(\text{availability} \times \text{role} \times \text{injury}\big), \quad (1)$$

so that 50 = an average NBA minute, 60+ = a quality starter, 80+ = a franchise engine (the 2025–26 maximum is Nikola Jokić at 97.8). The full specification — every component, weight, and shrinkage rule, seventeen equations of it — is in Appendix A, and the historical variant used for the seven-season series (GRAVITY-RS) is defined there too. Nothing in the body of the paper requires it; every table can be read with the z-score guide above.

> **In plain terms:** GRAVITY = (how good per minute) × (how reliably he's on the floor), graded against the whole league, every season, by the same formula. No reputation inputs, no external rankings. The judgment lives in the formula's construction — which is published in full, applied identically to all 644 players, and stress-tested in Appendix B.

3 College, 2017–2019: what the numbers saw and what they missed

Morant arrived at Murray State as an unranked three-star recruit with no high-major offers, famously spotted by an assistant coach in a side gym at a July 2016 camp. Two years later he was a consensus All-American and the #2 pick.

Table 1: Morant at Murray State. The jump to watch is the assist percentage — the share of teammate baskets he created — from 33.1% to an absurd 51.8%.

Season	G	MPG	PTS	REB	AST	TS%	3P%	FT%	USG%	AST%	BPM
2017–18 (FR)	32	34.0	12.7	6.5	6.3	.569	.307	.806	20.4	33.1	5.2
2018–19 (SO)	33	36.6	24.5	5.7	10.0	.612	.363	.813	33.3	51.8	10.0

His sophomore year made him the first Division I player ever to average 20 points and 10 assists. Here is the interesting experiment: feed that season into the same college production model GRAVITY uses to grade modern draft classes, standardized against the 2026 first-round cohort. The model scores a prospect on college BPM, efficiency, usage, and win shares:

Table 2: College Morant vs. the 2026 first-round cohort, in cohort z-scores. The production blend sees an ordinary lottery pick. The column it ignores is the outlier.

	BPM	TS%	USG%	WS/40	AST%
Morant 2018–19	10.0	.612	33.3	.273	51.8
2026 cohort mean	11.6	.604	25.9	.211	20.9
z-score	−0.54	+0.21	+1.71	+1.47	**+3.53**
Production blend = 0.5(−0.54) + 0.2(0.21) + 0.15(1.71) + 0.15(1.47) = +**0.25** (nearly nothing)					

By raw production, prime-prospect Morant was *unremarkable* — his college BPM sits below the 2026 first-round average (Cameron Boozer alone posted 18.7). What the formula excludes is the assist rate: +3.5 standard deviations above the cohort, 8.8 points higher than anyone in it. And creation turns out to be the one skill that survived every disaster of his career (Section 6). The model would have missed Ja Morant, and the miss is instructive.

One scope note. The yardstick is the 2026 first-round class because that is the cohort the current model grades — a convenience benchmark, not the information set a 2019 draft model would have had. The lesson does not depend on the choice: what makes the comparison instructive is its *shape* (ordinary production blend, historic creation), and no plausible cohort puts a 51.8% assist rate anywhere but the far tail.

In plain terms: College Ja looked statistically ordinary for a lottery pick — except for the passing, which was a one-in-thousands outlier that the standard formula doesn't even count. Lesson learned: creation now belongs in the draft model.

4 The rise, 2019–2022

Table 3 is the paper's centerpiece: every component of Morant's game, every season, graded against that season's league. Figure 1 plots the combined rating.

Table 3: The component matrix. Every cell is a z-score against that season's league (0 = average, +1 = good, +2 = elite, +3 = historic). IMPACT is the weighted combination; Pct is his percentile among 1,000+ minute players. Bold marks career highs and lows.

Season	MP	Box	WinSh	Volume	Effic.	Creation	Load	Stocks	Rebound	Rim	**IMPACT**	Pct
2019–20	2074	0.10	−0.22	0.69	−0.20	1.96	1.62	−0.74	−0.75	0.58	+0.24	63
2020–21	2053	−0.13	−0.45	0.77	−0.83	1.84	1.67	−0.88	−0.78	**1.18**	+0.04	55
2021–22	1889	**2.06**	**1.18**	**2.55**	**0.25**	2.13	2.50	−0.31	−0.22	0.86	**+1.74**	**97**
2022–23	1948	1.93	0.82	2.15	−0.72	2.91	2.92	−0.47	−0.07	1.20	+1.59	94
2023–24	318	0.99	0.38	1.60	−0.28	3.02	2.45	−0.61	−0.31	1.19	+0.63	80
2024–25	1519	0.80	0.17	1.81	−0.38	2.16	2.39	−0.44	−0.65	1.11	+0.81	83
2025–26	569	−0.48	**−1.31**	1.30	**−1.75**	**3.19**	**2.90**	−0.47	**−0.89**	0.82	**−0.11**	**43**

Three things stand out. First, the leap was a *scoring* leap, not a passing one: from 2020–21 to 2021–22 his volume z jumps 0.77 → 2.55, while his creation was already elite (≈ +1.9) as a rookie. Morant didn't learn to create; he learned to convert volume. Second, the leap held up in the playoffs, where it counts: an **8.4 playoff BPM** across two rounds in 2022 (27.1 points, 9.8 rebounds, 8.0 assists per game against Minnesota and Golden State) — at age 22, one of the strongest postseason signals any young guard has posted. Third, even at the summit there was a crack, visible in the Efficiency column: at his career-peak usage in 2022–23 (98th percentile of the league), his efficiency-for-that-burden was already *negative* (−0.72). The engine was extraordinary. The fuel economy never was.

In plain terms: 2021–22 Ja was a genuine top-10-caliber season (97th percentile), confirmed in the playoffs. But even peak Ja scored a lot without scoring *efficiently* — a flaw that was survivable at 22 and became the whole story at 26.

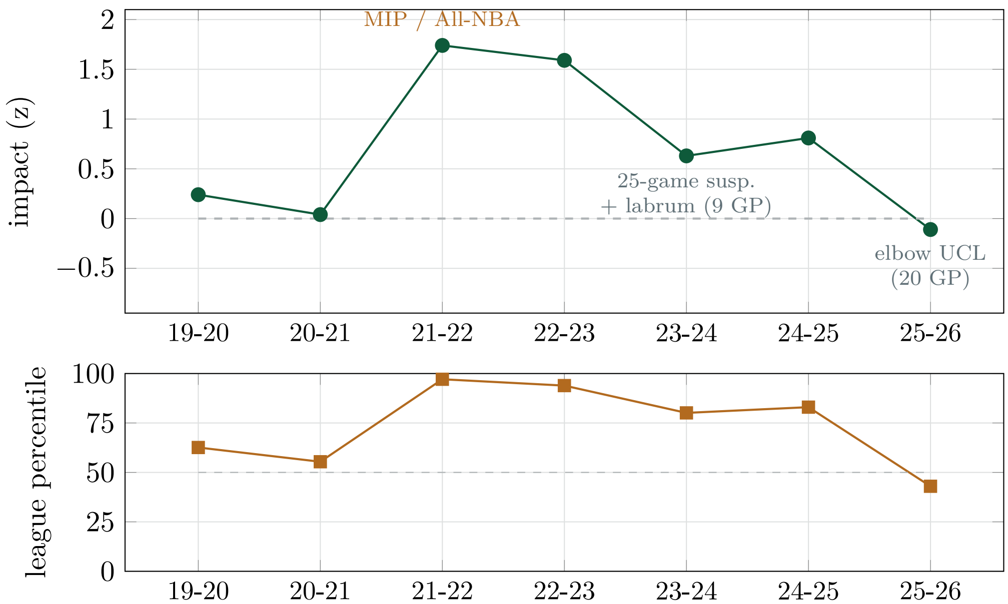

Figure 1: The arc. Top: per-season impact in z units (dashed line = league average). Bottom: the same thing as a percentile — in 2021–22 he was better than 97% of the league's rotation players; in 2025–26, better than 43%.

Table 4: The absence ledger. Across the three seasons 2023–24 to 2025–26, Morant appeared in 79 of a possible 246 regular-season games — 32.1%. The 2023–24 season accounts for 73 of the misses: the 25-game suspension first, then a nine-game return, then 48 games to the shoulder.

Date	Event	Consequence	Games lost
Mar 4, 2023	Firearm on Instagram Live (Denver)	League suspension	8
May 13, 2023	Second firearm video	League suspension (start of 23–24)	25
Jan 2024	Shoulder labrum tear (practice)	Surgery; season over after 9 GP	48
Nov 2025	"Go ask the coach" episode	Team suspension	1
Dec 2025	Right calf soreness	Two-week absence	∼7
Jan 21, 2026	Left elbow ligament sprain (vs. ATL)	Season over	33
Jun 29, 2026	Traded to Portland for Grant & Murray	—	—

5 The interruption years, 2023–2025

Then the availability collapsed. Table 4 is the ledger.

Two details in this stretch resist the standard telling. His nine-game 2023–24 cameo was *good* (25.1 points, 8.1 assists per game, 80th-percentile impact) before the shoulder ended it. And 2024–25, the forgotten season, was genuinely solid: over 1,519 minutes he graded at the 83rd percentile — roughly a top-50 player when on the floor — with a 94th-percentile foul-drawing rate. Morant did not slide gently for three years. He was a clearly good player right up until 2025–26, and then he fell off a ledge in one season.

6 The collapse season, 2025–26: three answers

Twenty games, 569 minutes: 19.5 points on .410 shooting, 8.1 assists, and the worst efficiency of his life. The season ended January 21 with an elbow ligament sprain, two games after he returned from a calf absence. We ask three questions of it.

6.1 Question 1: did his whole game collapse?

No — and this is the strangest fact in the paper. Compare the last two rows of Table 3, drawn in Figure 2. Every efficiency-linked number cratered to career-worst levels. And at the same time his **creation (+3.19) and offensive load (+2.90) hit career highs** — his assist rate (44.7%) was a 96th-percentile figure. The player in those 20 games was a career-best distributor carrying a career-peak burden who could not put the ball in the basket. That combination — process intact, outcomes broken — is what a cold streak looks like. Twenty games cannot prove that reading; what they can do is fail to match the alternative, because eroded athletes do not usually set career highs in the two most demanding process stats while their shots rim out.

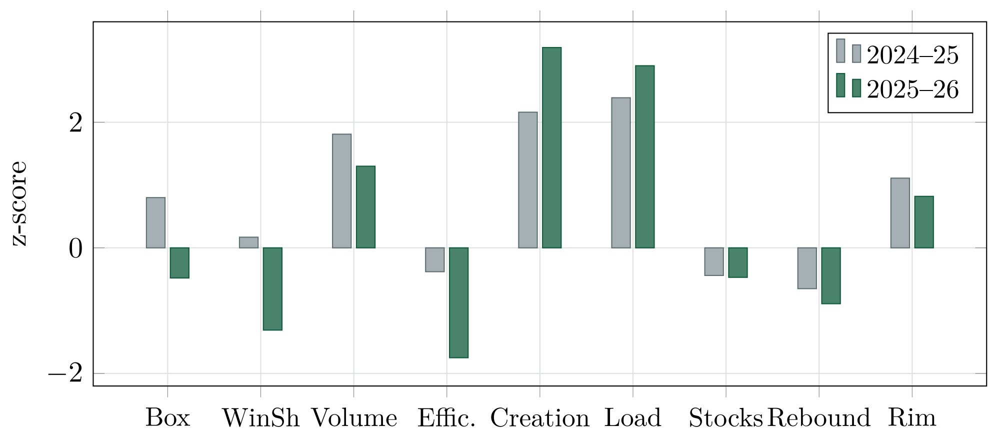

Figure 2: The paradox season. Green bars (2025–26) vs. grey (2024–25): the two tallest green bars — creation and load — are career highs; the efficiency bars are career lows.

6.2 Question 2: bad shots, or missed shots?

The visible symptom was an ugly shot diet: 46% of his attempts from mid-range, long twos tripling. Everyone blamed the shot selection. The accounting disagrees. Using CTG zone data (where shots came from, and how often they went in), the change in effective field-goal percentage between 2024–25 and 2025–26 splits *exactly* into a shot-selection part and a shot-making part:

$$\Delta\text{eFG} = \underbrace{\textstyle\sum_z (f'_z - f_z)\,\bar{a}_z\,w_z}_{\text{selection (where)}} + \underbrace{\textstyle\sum_z \bar{f}_z\,(a'_z - a_z)\,w_z}_{\text{making (whether)}}, \tag{2}$$

where f_z is the share of shots from zone z, a_z the accuracy there, and threes get their extra 50% credit (w_z). The split is symmetric — neither part is favored.

At the zone level, 88% of the collapse was missing, not choosing. Had he converted his ugly new shot diet at his normal rates, his efficiency would have been almost fine (.505). Fixing the diet without fixing the makes recovers almost nothing (.462). The shot chart looked broken because the shots missed, not mainly because of where they were taken.

Table 5: Where vs. whether, 2024–25 → 2025–26. His eFG fell about 5.7 points; selection explains 0.7 of them.

	Rim	Short mid	Long mid	Corner 3	Non-corner 3	eFG
Share of shots (24–25)	33%	35%	4%	6%	21%	.510
Accuracy (24–25)	62%	45%	39%	35%	32%	
Share of shots (25–26)	30%	37%	9%	4%	20%	.453
Accuracy (25–26)	61%	40%	39%	50%	19%	
Total drop = −5.7 pts = selection **−0.7** + making **−5.0**						
If he'd kept the *old diet* at the new accuracy: eFG = .462 (barely helps)						
If he'd hit the *new diet* at the old accuracy: eFG = .505 (nearly normal)						

One honest limitation. "Selection" here means *zone mix* — the five CTG zones — because that is what the decomposition can see. It cannot see defender distance, pull-up versus catch-and-shoot, shot-clock pressure, or how contested the attempts were; two non-corner threes can differ enormously in difficulty. If his 2025–26 attempts were systematically harder *within* zones, some of the "missing" is really hidden shot selection. The defensible reading is therefore: the collapse was not primarily about *where* he shot. How much of the conversion drop was shot difficulty rather than pure misses, zone data cannot separate — which is one more reason the sample-size analysis of the next section matters.

6.3 Question 3: is the missing even real?

Missing, then — over what sample? A 95% confidence interval answers the question every hot take skips: *given this few attempts, where could his true shooting level plausibly be?* (Formally we use Wilson score intervals; the formula is Appendix A, eq. 13.)

Table 6: 2025–26 shooting splits with 95% confidence bands. Read the last column: only one split moved beyond doubt, and it moved *up*.

Split	Made/Att	Season	Plausible range	Career	Verdict
3P%	20/85	.235	[.158, .336]	≈.315	career norm inside the band
FG%	132/322	.410	[.358, .464]	≈.465	borderline
Rim FG%	48/73	.658	[.544, .756]	≈.62	fully consistent
FT%	105/117	.897	[.829, .940]	.776	**significantly better**

The three-point collapse — the number most responsible for the "he's done" narrative — rests on 85 attempts, and its band comfortably contains his career norm. To be precise about what that means: the sample cannot confidently distinguish a genuine decline from ordinary shooting variance. It does not prove the miss was a cold streak; it proves nobody can yet prove it wasn't. Meanwhile the free throw, the purest touch test in basketball, improved so much the band excludes his career average from *below*. That is strong evidence against a broad loss of shooting touch — though not against every physical story, since a player can lose lift or separation on live-play jumpers while improving from a standstill. The simplest versions of "his shot is gone," however, predict the opposite of what happened at the stripe.

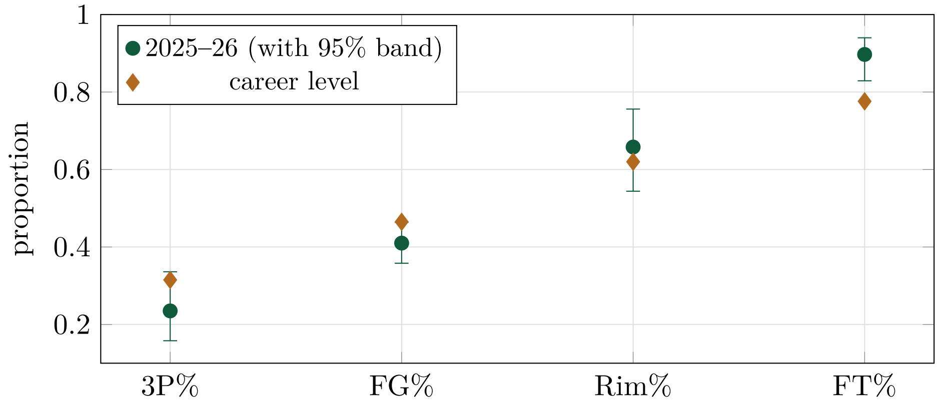

Figure 3: The sample-size problem, drawn. Green dots: 2025–26, with the range his true level could plausibly occupy. Amber diamonds: career levels. Three of four bands contain his career level. The one clear change is the free throw — upward.

In plain terms: Three answers: (1) his playmaking hit a career high while the shooting failed; (2) at the zone level, the failure was 88% missing, 12% shot selection; (3) at 85 threes, the sample can't tell genuine decline from an ordinary cold streak — and his free-throw touch got *better*. The one caveat that is real: his rim-attack rate has been sliding for seven years (Figure 4).

6.4 The seven-season shot profile

Career context, and the one concession the optimistic reading must make. His rim frequency has eroded steadily since the rookie peak — from 49% of attempts to 30% (Figure 4) — a real, slow trend consistent with age and injury. But the accuracy panel (Figure 5) shows what actually distinguished the collapse year: the three-ball fell out of its seven-year band, on the smallest sample, while rim conversion held steady and the free throw kept climbing.

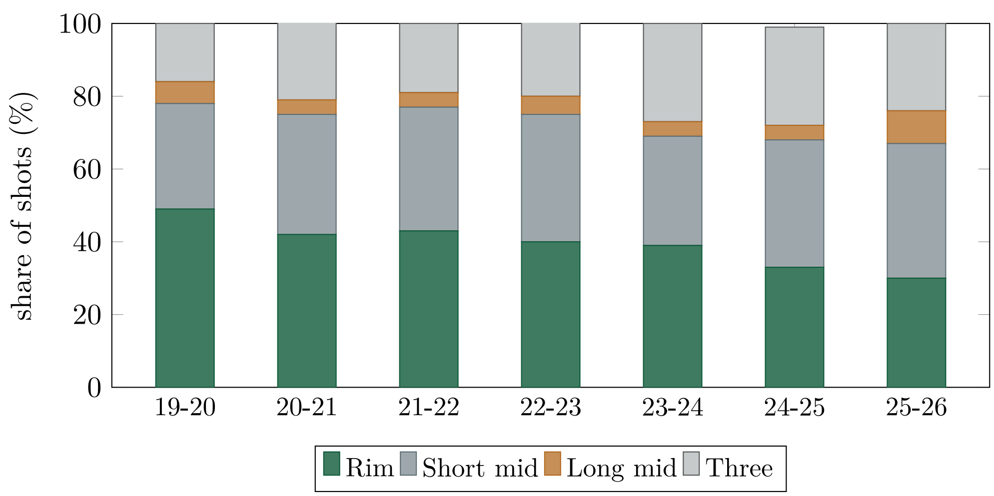

Figure 4: Where his shots come from, by season. The green (rim) share halves across seven years — the one long-run decline in the data that is unambiguously real.

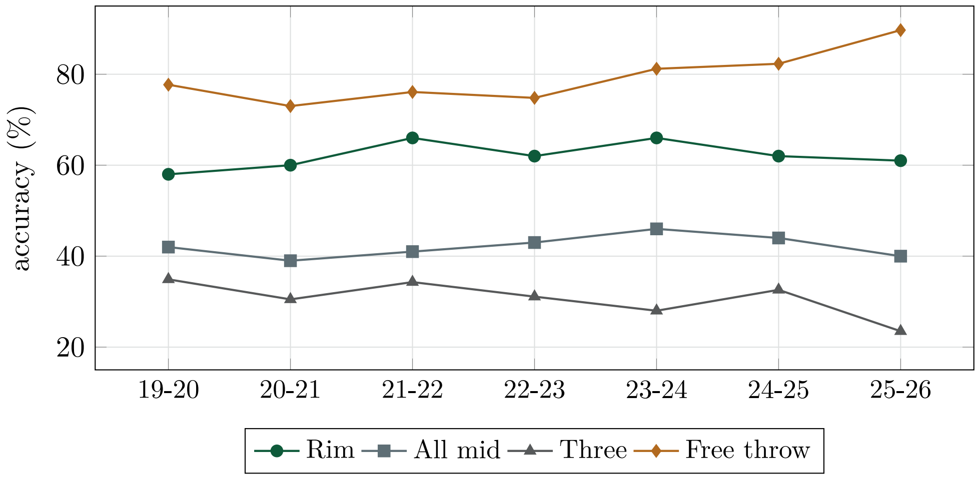

Figure 5: Whether they go in, by season. Rim conversion (green) holds a seven-year band; the free throw (amber) climbs to .897; the three (grey) crashes — on 85 attempts. The two purest touch measures moving in opposite directions is the season's central oddity, and randomness is its cheapest explanation.

7 How #140 happens

Table 7 runs Morant through the full GRAVITY pipeline, step by step (equation numbers refer to Appendix A). Notice what the arithmetic does and doesn't say. His blended two-season rate (+0.30) still describes an above-average NBA player. What produces #140 is the reliability stack: availability 0.896 (29 weighted games against a 55-game standard), role 0.958, injury override 0.950 — multiplying to 0.815 of his value above replacement. GRAVITY prices Morant like an asset with a high expected return and terrible delivery risk. Four days after the ranking date, an actual front office priced him identically, trading him for two salary-matching forwards. Because several steps in the pipeline are judgment calls — the override, the anchor, the season weights — Appendix B re-runs the entire ranking under alternative choices and reports how much #140 moves.

Table 7: The GRAVITY pipeline for Morant, July 2026, every step shown.

Step	Quantity	Value
Impact, 2025–26 [eq. 7]	569 minutes, on/off +1.5	−0.094
Impact, 2024–25 [eq. 7]	1519 minutes	+0.751
Blend weight [eq. 8]	recent season counts 0.75× per minute	0.529
Blended rate	$0.529(-0.094) + 0.471(0.751)$	+0.304
Small-sample shrink [eq. 9]	$c = \sqrt{2088/2200}$ toward replacement	+0.248
Playoff / age terms	—	0, 0
Availability [eq. 10]	29 weighted games of 55	× 0.896
Role size [eq. 11]	28.4 min/game	× 0.958
Injury override [eq. 12]	elbow ligament, season-ending	× 0.950
Final z	$-1.9 + (0.248 + 1.9)(0.815)$	−0.149
gravity score	$50 + 15(-0.149)$	**47.8**
Rank	of 644 (all veterans + 2026 draft class)	**#140**

8 What Portland is getting: the price of every future

Morant enters 2026–27 at 27, sharing a backcourt with Damian Lillard (36, returning from an Achilles rupture), on a roster that just made the playoffs without either of them. His future has two separate uncertainties — *how well he plays* and *how often he plays* — and bundling them into named scenarios hides the trade. So Table 8 prices them separately: each cell runs the full pipeline (blend with the 569-minute 2025–26 season, credibility shrink, availability and role multipliers) for an assumed 2026–27 per-minute impact and games total, at 31 minutes per game with no injury override assumed. These are *illustrative prices, not probabilities*: the framework says what each future is worth, not which one arrives.

Table 8: The availability × impact matrix. Cell = projected July 2027 score (equivalent rank in the July 2026 edition). Row levels are his own career benchmarks: −0.10 is the 2025–26 collapse, +0.80 is 2024–25, +1.65 is the 2021–23 peak.

2026–27 per-minute level	25 GP	50 GP	65 GP
−0.10 (repeat collapse)	39.7 (#281)	46.2 (#166)	47.8 (#138)
+0.80 (2024–25 level)	47.1 (#154)	57.2 (#53)	59.8 (#35)
+1.65 (2021–23 peak)	54.0 (#77)	67.6 (#10)	71.2 (#7)

The matrix makes three facts legible that the bundled scenarios obscured. First, the corner cells agree on a striking symmetry: playing at his 2024–25 level for only 25 games (47.1) is worth almost exactly the same as playing a full healthy season at his collapse level (47.8) — and the latter is his current score. Health and form are near-perfect substitutes at the bottom of the table. Second, the big jumps live in the *games* direction: at the 2024–25 level, going from 25 to 50 games buys +10.1 points; going from the 2024–25 level to the peak level at fixed 50 games buys +10.4. Portland's return on a healthy Morant is as large as its return on a resurgent one. Third, top-50 requires *both*: no 25-game column reaches it, and no collapse-level row does.

Which cells deserve weight is a judgment the reader can make with the paper's evidence: the skills that collapse first in genuinely declining guards — rim conversion, foul drawing, free-throw touch — are respectively stable, elite (seven straight seasons at the 86th percentile or better), and improving, while the stats that cratered are the ones a confidence interval refuses to certify. That argues against loading the bottom row. Against it stand the two real risks: a rim-attack rate that has halved in seven years, and the availability record of Table 4. The binding constraint on Ja Morant has never been a shooting slump. It is that since 2023 he has played 32.1% of his team's games. GRAVITY will believe the rise again when the denominator does.

9 Conclusion

Measured with one instrument across seven seasons, Morant's career is a steep genuine rise ($+1.7\sigma$, twice), an availability catastrophe, and one 569-minute season the data cannot fully judge. Most of the observed efficiency decline came from worse conversion rather than from where he shot — though the sample is too small to say how much of that conversion drop is lasting decline rather than variance. What the data can certify: his creation set a career high, his rim conversion held, his free-throw touch improved; against that, his rim-attack rate has fallen for seven straight years and his availability record is severe. #140 of 644 is a defensible price for that bundle of uncertainty — robust, per Appendix B, to every judgment call we could vary — and it should not be read as a claim that 139 players possess more healthy, on-court talent. It is the right price for the risk, and probably the wrong forecast of the talent. Both statements are the same mathematics, read in two directions.

Revision note (v2, July 11, 2026)

Following a detailed external review, this version corrects two accounting errors and tightens several claims. (1) The absence ledger charged 73 games to the January 2024 shoulder injury; the correct figure is 48, since the 25-game suspension — listed on its own row — and the nine games he played account for the rest of that season. (2) The three-season availability denominator was stated as 249 games (31.7%); the correct figures are 246 and 32.1%. (3) The Shapley decomposition is now explicitly scoped to zone-level selection; the free-throw and cold-streak inferences are stated with the caveats the intervals actually support; the three bundled forward scenarios were replaced by the availability × impact matrix of Table 8, recomputed under a single stated convention; and Appendix B (model sensitivity) was added. The errors were ours. Nothing in the correction changes the ranking itself, which is computed from the data, not from the prose.

A The full GRAVITY specification

For completeness; nothing in the body requires this section. Let season y have player set $\mathcal{P}_y$ with minutes m_i. For any statistic $x_k(i)$, minutes-weighted moments and clipped z-scores are

$$\mu_k = \frac{\sum_i m_i x_k(i)}{\sum_i m_i}, \quad \sigma_k = \left(\frac{\sum_i m_i (x_k(i)-\mu_k)^2}{\sum_i m_i}\right)^{1/2}, \quad z_k(i) = \mathrm{clip}\big((x_k(i)-\mu_k)/\sigma_k,\ \pm 4\big). \tag{3}$$

Component constructions:

$$\mathrm{eff}(i) = (\mathrm{TS}_i - \mu_{\mathrm{TS}}) \cdot 100 \cdot (\mathrm{clip}(\mathrm{USG}_i, 8, 36)/20)^{0.7}, \tag{4}$$

$$\mathrm{creation}(i) = \mathrm{AST\%}_i - 0.55\,\mathrm{TOV\%}_i, \qquad \mathrm{load}(i) = \mathrm{USG}_i + 0.4\,\mathrm{AST\%}_i, \tag{5}$$

$$\mathrm{stocks}(i) = 2\,\mathrm{STL\%}_i + 1.2\,\mathrm{BLK\%}_i, \qquad \mathrm{rim}(i) = 100\,\mathrm{FTr}_i + 200 \cdot \text{dunk share}_i. \tag{6}$$

Skill blend (current season): $S = 0.27 z_{\mathrm{vol}} + 0.19 z_{\mathrm{eff}} + 0.22 z_{\mathrm{crea}} + 0.09 z_{\mathrm{load}} + 0.10 z_{\mathrm{stk}} + 0.05 z_{\mathrm{reb}} + 0.08 z_{\mathrm{rim}}$. Season impact, with CTG filtered on/off regressed by minutes and a within-season small-sample shrink:

$$I_y(i) = \Big[0.44 z_{\mathrm{BPM}} + 0.12 z_{\mathrm{WS/48}} + 0.14\, z\big(\Delta_{\mathrm{on/off}} \min(1, m_i/2400)\big) + 0.30\, S\Big] \sqrt{\min(1, m_i/800)}. \tag{7}$$

Two-year blend and credibility shrink toward replacement $\rho = -1.9$:

$$w = \frac{0.75 m_y}{0.75 m_y + 0.25 m_{y-1}}, \qquad R = w I_y + (1-w) I_{y-1}, \tag{8}$$

$$\tilde{R} = cR + (1-c)\rho, \qquad c = \sqrt{\frac{\min(m_y + m_{y-1}, 2200)}{2200}}. \tag{9}$$

Playoff evidence $\pi = \mathrm{clip}(0.12(\mathrm{BPM}^{po} - \mathrm{BPM}^{rs}) \min(1, m^{po}/300)/2.6, \pm 0.30)$; age nudge α (+0.055/yr under 24, −0.045/yr past 30, clipped $[-0.30, 0.25]$). Reliability multipliers:

$$A = 0.78 + 0.22 \min\big(1, (0.7 g_y + 0.3 g_{y-1})/55\big), \tag{10}$$

$$\Phi = 0.62 + 0.38 \min\big(1, \max(\mathrm{mpg}_y, 0.85\,\mathrm{mpg}_{y-1})/32\big), \tag{11}$$

$$J \in [0.78, 1.00] \text{ (documented injury override)}. \tag{12}$$

Final score: $G = 50 + 15[\rho + (\tilde{R} + \pi + \alpha - \rho)A\Phi J]$. The historical variant GRAVITY-RS used for Table 3 deletes the on/off term (not uniformly available pre-2024) and renormalizes: $I^{RS} = [0.533z_{\text{BPM}} + 0.133z_{\text{WS/48}} + 0.333S']\sqrt{\min(1, m/800)}$, with S' using prior-season skill weights and a free-throw-rate-only rim term. Wilson 95% intervals used in Section 6.3:

$$p \in \left(\hat{p} + \tfrac{z^2}{2n} \pm z\sqrt{\tfrac{\hat{p}(1-\hat{p})}{n} + \tfrac{z^2}{4n^2}}\right) \Big/ \left(1 + z^2/n\right), \qquad z = 1.96. \tag{13}$$

Data

Basketball-Reference league, player and playoff tables (2019–20 to 2025–26); Cleaning the Glass subscriber database (filtered on/off, shot zones, foul drawing), accessed July 10, 2026; Sports-Reference CBB. The specification is identical to the July 10, 2026 "The 644" ranking (v3), in which Morant's pipeline values appear verbatim. No external rankings or projections were consulted.

B How much do the model's choices matter?

A transparent model is not the same thing as a validated one. The weights in Appendix A — and the replacement anchor, the availability standard, and the injury overrides — were specified by judgment before any player was ranked; they were not fitted to a target variable. Publishing them makes the judgment inspectable. It does not make it correct. The honest next question is: how much does this paper's conclusion depend on those choices?

Table 9 answers it by re-running the *entire* 644-player pipeline — rookies included, since their scores are percentiles of the veteran distribution — under each variation, and reporting where Morant lands. Each row changes exactly one thing.

Table 9: Sensitivity of Morant's July 2026 placement. Every variation re-ranks all 644 players; rank is within that variation's own ranking.

Model variation	Score	Rank
Baseline GRAVITY v3, as published	47.8	#140
Injury overrides removed entirely (all J=1)	49.1	#130
Equal two-season weights (0.50/0.50 vs. 0.75/0.25)	50.3	#112
Availability standard 65 games (vs. 55)	47.2	#147
Alternate impact weights (BPM .35, WS/48 .15, on/off .20)	47.5	#147
Replacement level −1.6 (vs. −1.9)	48.7	#138
Reliability stack removed (A=Φ=J=1; per-minute talent only)	53.7	#112

Every variation lands him between #112 and #147. The two that move him most — equal season weights, and deleting the reliability machinery altogether — are both versions of the same question, *how much should the solid 2024–25 season count?*, and both answer #112: even the reading most generous to his track record does not put him inside the top 100 on this data. In the other direction, no tightening we tested pushes him past #147. The paper's headline number is not an artifact of any single judgment call; within this family of specifications, "a clearly good player, heavily discounted for delivery risk" is the only available conclusion.

What this appendix cannot do is validate the model against the future — that is what the Forecast Ledger is for. GRAVITY's registered predictions are graded in public, annually, starting with the July 2027 Forecast Audit; predictive validation accrues there, one scored forecast at a time.

References

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